

Chapitre 10 : Interface utilisateur

Solutions exercices

Exercice 1

```
import java.awt.FlowLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JTextField;

public class Chap10Ex1 {

    private JFrame frame;
    private JLabel lblCel;
    private JLabel lblFah;
    private JTextField txtCel;
    private JButton btnConvertir;

    public Chap10Ex1() {
        frame = new JFrame();
        frame.setSize(300, 150);
        frame.setLocation(100, 100);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.getContentPane().setLayout(new FlowLayout());
        frame.setTitle("Celsius à Fahrenheit");

        lblCel = new JLabel("Température en Celsius:   ");
        frame.getContentPane().add(lblCel);

        txtCel = new JTextField();
        txtCel.setColumns(20);
        frame.getContentPane().add(txtCel);

        btnConvertir = new JButton("Convertir");
        btnConvertir.addActionListener(new BtnConvertirActionListener());
        frame.getContentPane().add(btnConvertir);

        lblFah = new JLabel("Température en Fahrenheit: --");
```

```

        frame.getContentPane().add(lblFah);
    }

    public static void main(String[] args) {
        Chap10Ex1 fenetre = new Chap10Ex1();
        fenetre.frame.setVisible(true);
    }

    private class BtnConvertirActionListener implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            double celsius;
            double fah;

            celsius = Double.parseDouble(txtCel.getText());
            fah = celsius * 1.8 + 32;
            lblFah.setText("Temperature en fahrenheit: " + fah);
        }
    }
}

```

Exercise 2

```

import java.awt.FlowLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JTextField;

public class Chap10Ex2 {
    private JFrame frame;
    private JLabel lblRayon;
    private JTextField txtRayon;
    private JButton btnCalculer;
    private JLabel lblAire;
    private JLabel lblCirc;
}

```

```

private JLabel lblDia;

public Chap10Ex2() {
    frame = new JFrame();
    frame.setSize(290, 210);
    frame.setLocation(100, 100);
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frame.getContentPane().setLayout(null);

    lblRayon = new JLabel("Rayon :");
    lblRayon.setSize(70, 15);
    lblRayon.setLocation(40, 20);
    frame.getContentPane().add(lblRayon);

    txtRayon = new JTextField();
    txtRayon.setSize(115, 25);
    txtRayon.setLocation(125, 20);
    frame.getContentPane().add(txtRayon);

    btnCalculer = new JButton("Calculer");
    btnCalculer.setSize(115, 25);
    btnCalculer.setLocation(125, 60);
    btnCalculer.addActionListener(new BtnCalculerActionListener());
    frame.getContentPane().add(btnCalculer);

    lblAire = new JLabel("Aire :");
    lblAire.setSize(310, 15);
    lblAire.setLocation(40, 100);
    frame.getContentPane().add(lblAire);

    lblCirc = new JLabel("Circonférence :");
    lblCirc.setSize(310, 15);
    lblCirc.setLocation(40, 120);
    frame.getContentPane().add(lblCirc);

    lblDia = new JLabel("Diamètre :");
    lblDia.setSize(310, 15);
    lblDia.setLocation(40, 140);
    frame.getContentPane().add(lblDia);
}

public static void main(String[] args) {
    Chap10Ex2 fenetre = new Chap10Ex2();
}

```

```

        fenetre.frame.setVisible(true);
    }

    private class BtnCalculerActionListener implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            Cercle monCercle = new Cercle();
            monCercle.setRayon(Double.parseDouble(txtRayon.getText()));
            lblAire.setText("Aire: " + monCercle.aire());
            lblCirc.setText("Circonférence: " + monCercle.circonference());
            lblDia.setText("Diamètre: " + monCercle.diametre());
        }
    }
}

```

```

public class Cercle {

    private double rayon;

    public Cercle() {
        rayon = 0;
    }

    public double getRayon() {
        return rayon;
    }

    public void setRayon(double rayon) {
        this.rayon = rayon;
    }

    double diametre() {
        return 2 * rayon;
    }

    double circonference(){
        return 2 * 3.1416 * rayon;
    }

    double aire(){
        return 3.1416 * rayon * rayon;
    }
}

```

```
}
```

Exercise 3

```
import java.awt.Color;
import java.awt.Graphics;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JPanel;
import javax.swing.JTextField;
import javax.swing.SwingConstants;
import javax.swing.border.LineBorder;

public class Chap10Ex3 {
    private JFrame frame;
    private JPanel panel;

    private JLabel lblHauteur;
    private JTextField txtHauteur;
    private JLabel lblPositionX;
    private JTextField txtPosX;
    private JLabel lblPositionY;
    private JTextField txtPosY;
    private JButton btnDessiner;

    private int hauteur = 0;
    private int posX = 0;
    private int posY = 0;

    public Chap10Ex3() {
        frame = new JFrame();
        frame.setSize(470, 380);
        frame.setLocation(0, 0);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.getContentPane().setLayout(null);

        panel = new JPanel();
```

```
panel.setSize(315, 330);
panel.setLocation(10,10);
panel.setBorder(new LineBorder(new Color(0, 0, 0)));
frame.getContentPane().add(panel);

lblHauteur = new JLabel("Hauteur");
lblHauteur.setSize(110, 20);
lblHauteur.setLocation(340, 10);
lblHauteur.setHorizontalAlignment(SwingConstants.CENTER);
frame.getContentPane().add(lblHauteur);

txtHauteur = new JTextField();
txtHauteur.setSize(110, 20);
txtHauteur.setLocation(340, 30);
frame.getContentPane().add(txtHauteur);

lblPositionX = new JLabel("Position X");
lblPositionX.setSize(110, 20);
lblPositionX.setLocation(340, 70);
lblPositionX.setHorizontalAlignment(SwingConstants.CENTER);
frame.getContentPane().add(lblPositionX);

txtPosx = new JTextField();
txtPosx.setSize(110, 20);
txtPosx.setLocation(340, 90);
frame.getContentPane().add(txtPosx);

lblPositionY = new JLabel("Position Y");
lblPositionY.setSize(110, 20);
lblPositionY.setLocation(340, 130);
lblPositionY.setHorizontalAlignment(SwingConstants.CENTER);
frame.getContentPane().add(lblPositionY);

txtPosy = new JTextField();
txtPosy.setSize(110, 20);
txtPosy.setLocation(340, 150);
frame.getContentPane().add(txtPosy);

btnDessiner = new JButton("Dessiner");
btnDessiner.addActionListener(new BtnDessinerActionListener());
btnDessiner.setSize(110, 30);
btnDessiner.setLocation(340, 210);
frame.getContentPane().add(btnDessiner);
```

```

    }

    public static void main(String[] args) {
        Chap10Ex3 fenetre = new Chap10Ex3();
        fenetre.frame.setVisible(true);
        fenetre.frame.repaint();
    }

    private class GPanel extends JPanel {
        @Override
        public void paintComponent(Graphics g) {
            super.paintComponent(g);

            g.setColor(Color.black);
            g.drawRect(posX, posY - hauteur/2, hauteur/2, hauteur/2);
            g.drawLine(posX, posY - hauteur/2, posX + hauteur/4, posY - hauteur);
            g.drawLine(posX + hauteur/4, posY - hauteur, posX + hauteur/2, posY - hauteur/2 );
        }
    }

    private class BtnDessinerActionListener implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            hauteur = Integer.parseInt(txtHauteur.getText());
            posX = Integer.parseInt(txtPosx.getText());
            posY = Integer.parseInt(txtPosy.getText());

            frame.repaint();
        }
    }
}

```

Exercice 4

```

public class Temperature {
    private double valeur;

    public Temperature() {
        valeur = 0;
    }
}

```

```
    public double getValeur() {  
        return valeur;  
    }  
  
    public void setValeur(double valeur) {  
        this.valeur = valeur;  
    }  
  
    double c2f(){  
        return (valeur * 9 / 5 + 32);  
    }  
  
    double f2c(){  
        return ((valeur - 32) * 5 / 9);  
    }  
}
```

```
import java.awt.EventQueue;  
  
import javax.swing.JFrame;  
import javax.swing.JLabel;  
import javax.swing.JTextField;  
import javax.swing.JButton;  
import javax.swing.JComboBox;  
import javax.swing.SwingConstants;  
  
import java.awt.event.ActionListener;  
import java.awt.event.ActionEvent;  
  
public class Chap10Ex4 {  
  
    private JFrame frmConversion;  
    private final JLabel lblIn;  
    private final JTextField textField;  
    private final JButton btnConvertir;  
    private final JLabel lblOutput;  
    private final JLabel label;  
    private final JComboBox comboBox;  
  
    public Chap10Ex4() {
```



```

    frmConversion = new JFrame();
    frmConversion.setTitle("Conversion");
    frmConversion.setBounds(100, 100, 245, 240);
    frmConversion.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frmConversion.getContentPane().setLayout(null);

    lblIn = new JLabel("Température en Celsius");
    lblIn.setHorizontalAlignment(SwingConstants.CENTER);
    lblIn.setBounds(0, 22, 233, 20);
    frmConversion.getContentPane().add(lblIn);

    textField = new JTextField();
    textField.setBounds(80, 53, 86, 20);
    textField.setColumns(10);
    frmConversion.getContentPane().add(textField);

    btnConvertir = new JButton("Convertir");
    btnConvertir.setBounds(66, 85, 118, 23);
    btnConvertir.addActionListener(new BtnConvertirActionListener());
    frmConversion.getContentPane().add(btnConvertir);

    lblOutput = new JLabel("Température en Fahrenheit");
    lblOutput.setHorizontalAlignment(SwingConstants.CENTER);
    lblOutput.setBounds(12, 119, 221, 14);
    frmConversion.getContentPane().add(lblOutput);

    label = new JLabel("- - -");
    label.setHorizontalAlignment(SwingConstants.CENTER);
    label.setBounds(0, 145, 233, 14);
    frmConversion.getContentPane().add(label);

    String choix[] = {"C --> F", "F --> C"};
    comboBox = new JComboBox(choix);
    comboBox.setBounds(80, 171, 86, 20);
    comboBox.addActionListener(new ComboBoxActionListener());
    frmConversion.getContentPane().add(comboBox);
}

public static void main(String[] args) {

    Chap10Ex4 window = new Chap10Ex4();
    window.frmConversion.setVisible(true);
}

```

```

    }

    private class BtnConvertirActionListener implements ActionListener {
        public void actionPerformed(ActionEvent arg0) {
            Temperature maTemp = new Temperature();
            maTemp.setValeur(Double.parseDouble(textField.getText()));
            if (comboBox.getSelectedIndex() == 0) {

                label.setText(maTemp.c2f() + " C");
            }
            else {
                label.setText(maTemp.f2c() + " F");
            }
        }
    }

    private class ComboBoxActionListener implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            if (comboBox.getSelectedIndex() == 0) {

                lblIn.setText("Température en Celsius");
                lblOutput.setText("Temp\u00E9rature en Fahrenheit");
            }
            else {
                lblIn.setText("Temp\u00E9rature en Fahrenheit");
                lblOutput.setText("Température en Celsius");
            }
        }
    }
}

```